TASK 1

Research Question: Are product orders uniformly distributed across the 21 high-level departments?

DEFINING HYPOTHESES

H_0: Orders are uniformly distributed

H_a: At least one department's proportion differs from 1/21

DETERMINING DEGREES OF FREEDOM

There are k=21 departments

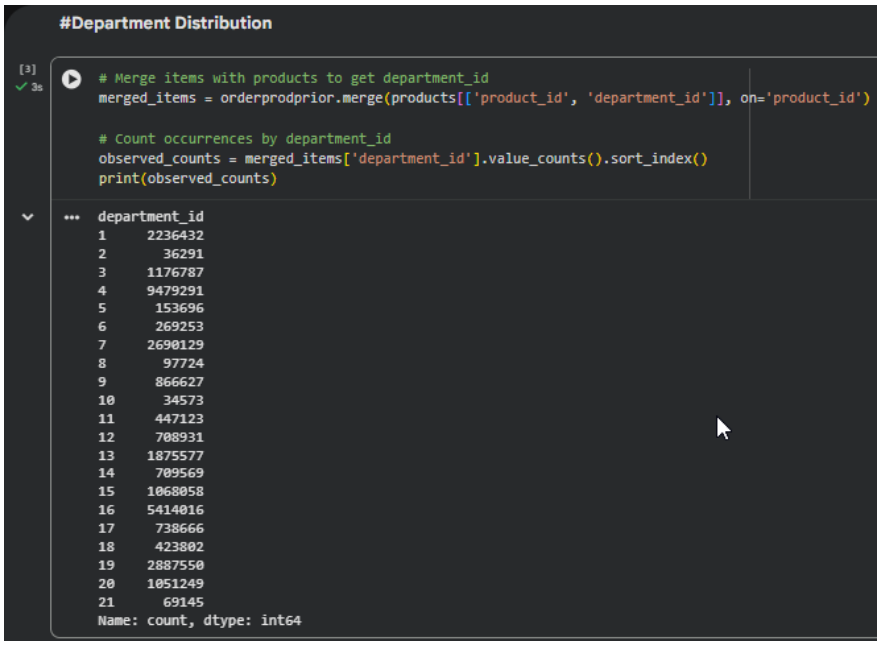
df = k – 1 = 21 -1 = 20

df= 20

EXPECTED FREQUENCIES

E = n/21

1.Observed frequencies



2. Equal Proportions

Total items(n): 32,434,488

Number of expected categories(k): 21

E= 32,434,488/21= 1,544,499.42 items per department

Under H_0 , we expect approximately 1.54 million items to be sold in each department.

3.Goodness of fit

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Critical value: At a 5% level of significance (a=0.05) and df=20, the benchmark is 31.41.

Working for Top Drivers(Sample):

- **Department 4 (Produce):** $\frac{(9{,}479{,}291 - 1{,}544{,}499.48)^2}{1{,}544{,}499.48} = 40{,}758{,}260.6$
- **Department 16 (Dairy Eggs):** $\frac{(\mathbf{5{,}414{,}016 - 1{,}544{,}499.48})^2}{\mathbf{1{,}44{,}499.48}}$ = 9,694,763.5

Conclusion: Because the calculated statistic (the sum of all 21 components) will be in the tens of millions, it far exceeds the critical value of 31.41. Therefore, we reject the null hypothesis (H_0).

4. Categories contributing most to χ2

The Primary Contributors are those with the largest squared deviations from the expected mean of 1.54 million.

1. Produce (Dept 4): With over 9.4 million items, this is the single largest driver of the non-uniformity, as it sells six times the expected volume.
2. Dairy Eggs (Dept 16): With 5.4 million items, it significantly exceeds the expected uniform share.
3. Low-Volume Departments (e.g Dept 2, 10, 21): These departments (Bulk, Other, Personal Care) have very low counts (O<70,000). Because their actual counts are

so much smaller than the 1.54 million expected, their squared deviations also contribute heavily to the total statistic.

Operational Interpretation for Management

This analysis proves that consumer behaviour on Instacart is highly concentrated. Management should:

- Asymmetric Labor Allocation: Focus the majority of picking and warehouse staff on the Produce and Dairy aisles.
- Inventory Velocity: Implement high-frequency restocking for high-volume drivers while reducing the footprint for low-contribution areas like Bulk (Dept 2) or Personal Care (Dept 10).

Task 2: Hour-of-Day Distribution Investigation

Hypotheses and Parameters

H_0: Ordering activity is uniformly distributed across the day (p_0 = p_1 = …= p_{23}= 1/24).

H_a: Ordering activity is not uniform; at least one hour's proportion differs from 1/24.

Degrees of freedom(df): k-1 = 24-1= 23

Critical Value: At a significance level of 0.05 and df 23, the rejection benchmark is 35.172.

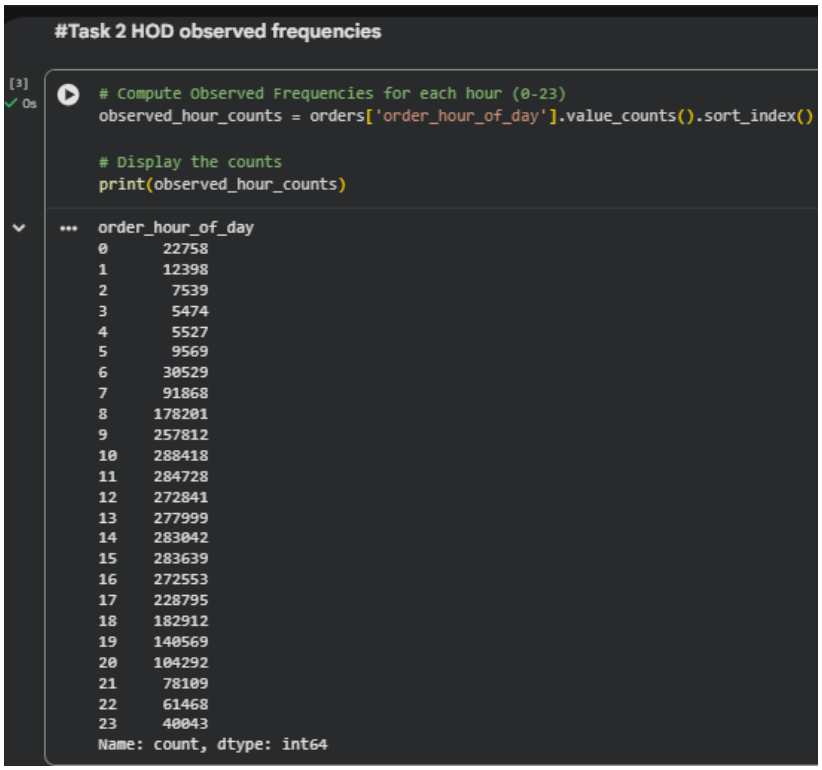
Expected Frequencies

E= n/24

E= 3,421,083/24 = 142,545.125 orders per hour

Validation: All $E_i \geq 5$, satisfying the "Rule of Five" for a valid Chi-Square test.

Observed frequencies(O)



#Task 2 HOD observed frequencies

```
[3]
✓ 0s
# Compute Observed Frequencies for each hour (0-23)
observed_hour_counts = orders['order_hour_of_day'].value_counts().sort_index()

# Display the counts
print(observed_hour_counts)
```

```
order_hour_of_day
0      22758
1      12398
2       7539
3       5474
4       5527
5       9569
6      30529
7      91868
8     178201
9     257812
10    288418
11    284728
12    272841
13    277999
14    283042
15    283639
16    272553
17    228795
18    182912
19    140569
20    104292
21     78109
22     61468
23     40043
Name: count, dtype: int64
```

Counts per Hour

HOUR	COUNT(O)	HOUR	COUNT(O)	HOUR	COUNT(O)	HOUR	COUNT(O)
0	22,758	6	30,529	12	272,841	18	182,912
1	12,398	7	91,868	13	277,999	19	140,569
2	7,539	8	178,201	14	283,042	20	104,292
3	5,474	9	257,812	15	283,639	21	78,109
4	5,527	10	288,418	16	272,553	22	61,468
5	9,569	11	284,728	17	228,795	23	40,043

GOF TEST

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$ is the formula used

For Key deviations:

- **Peak Hour (10:00 AM):** $\frac{(288{,}418 - 142{,}545.125)^2}{142{,}545.125}$ = 149,307.7
- **Lowest Hour (3:00 AM):** $\frac{(5{,}474 - 142{,}545.125)^2}{142{,}545.125}$ = 131,811.5

Conclusion: When summing all 24 hourly components, the total Calculated Test Statistic will be in the millions, which is vastly greater than the critical value of **35.172**. Therefore, we **Reject the Null Hypothesis.**

Identifying Categories contributing most to χ2

The "Primary Drivers" of this rejection are the hours with the most extreme deviations from the average of 142,545 orders:

- **The Midday Plateau (Hours 9–16):** These hours consistently double the expected frequency, representing the primary positive contributors to the statistic.
- **Early Morning Dormancy (Hours 1–5):** These hours show almost zero activity relative to the expectation, contributing massive squared deviations.

Operational Interpretation for Management

This statistical evidence confirms that Instacart's demand is highly non-uniform, necessitating the following business strategies:

Dynamic Staffing: Shift the majority of fulfillment labor and customer support to the 10:00 AM – 4:00 PM window to handle the midday rush.

Maintenance Windows: The period between 1:00 AM and 5:00 AM is "statistically silent." Management should strictly schedule all system updates and warehouse restocking during this window to minimize customer disruption.

Demand Shifting: Prescriptive analytics could be used to offer "early bird" or "late night" delivery discounts to shift volume away from the peak midday hours and balance the load.

Task 3: Reorder Flag Distribution Investigation

1.Hypotheses and Parameters

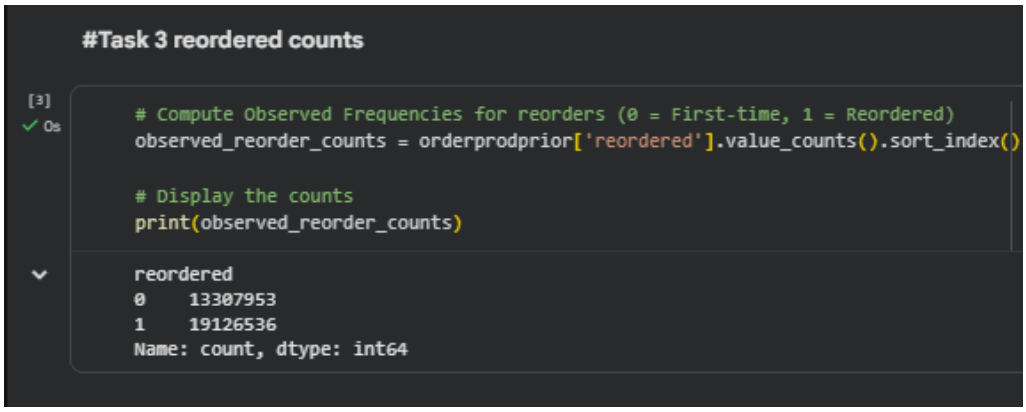
Null Hypothesis: Reorders and first-time orders follow a 50–50 distribution (preordered= 0.5; $p_{first_time} = 0.5$)

Alternative Hypothesis: The distribution of orders is not 50–50; one category significantly outweighs the other.

Degreed of freedom(df): k – 1 = 2-1= 1

Critical Value: At a 5% level of significance and df=1, the benchmark is 3.841

2. Observed and Expected Frequencies



- Habitual Dominance: Approximately 59% of all items added to carts are reorders. This suggests that "habit-formation" is the primary driver of the instacart business model.

Observe Frequencies(O):

- First time orders (0): 13,307,953
- Reordered items (1): 19,126,536

Expected Frequencies(E):

32,434,489 x 0.5 = 16,217,244.5 items for each category

3.GOF test

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$ is the formula used

Category	Observes(O)	Expected(E)	Deviation(O-E)	Squared Deviation $(O-E)^2$	Component $\frac{(O-E)^2}{E}$
First time(0)	13,307,953	16,217,244.5	-2,909,291.5	8.4639e+12	521,912.1

Reorderer(1)	19,126,536	16,217,244.5	2,909,291.5	8.4639e+12	521,912.1
TOTAL	32,434,489	32,434,489	0		1,043,824.2

The total Calculated Test Statistic is approximately 1,043,824.2

4.Statistical Conclusion

Decision Rule: Reject H_0 if χ2 calc> 3.841

Comparison: 1,043,824.2>3.841

Conclusion: We Reject the Null Hypothesis (H_0). There is overwhelming statistical evidence to conclude that the distribution of orders is not balanced 50–50.